“Bad statistics makes bad research, and bad research is not ethical”

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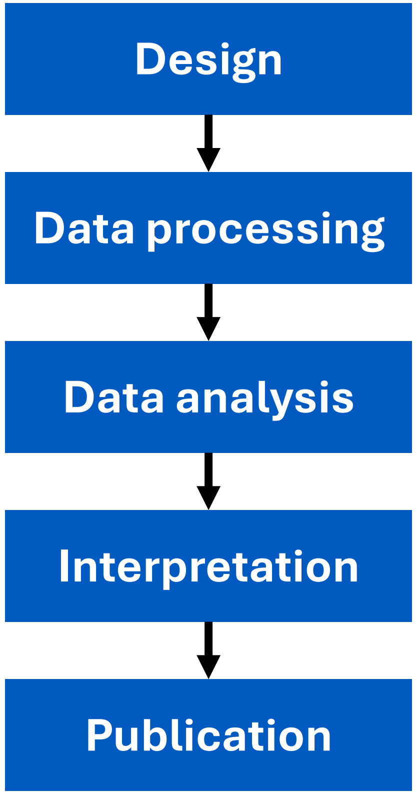
- Statistical errors may occur at all stages of a study
- One error may render a whole study useless
- Publishing incorrect or misleading findings:

1. Affects patient care
2. Constitutes research waste
3. Causes further research waste

- If bad statistics is unethical, then why does it happen?

85% of $100bn a year spent on clinical trials globally is wasted avoidably



Chalmers I, Glasziou P: Avoidable waste in the production and reporting of research evidence. *The Lancet 2009, 374(9683):86–89*

“We need less research, better research, and research done for the right reasons”

- Not much improvement since 1994
- You need to publish to advance your career, so what can you do?

“Do the best you can until you know better. Then when you know better, do better” – Maya Angelou

“Start thinking like a medical statistician” – Me

Publish or perish

+

Poor statistical training

+

Lack of statisticians

=

Research system failing researchers and patients

Altman DG: The scandal of poor medical research. *BMJ 1994, 308(6924)*

BE SCEPTICAL

Scepticism makes science self-correcting

Statistics is a science, not a tool

Statistics is hard, so copying is common

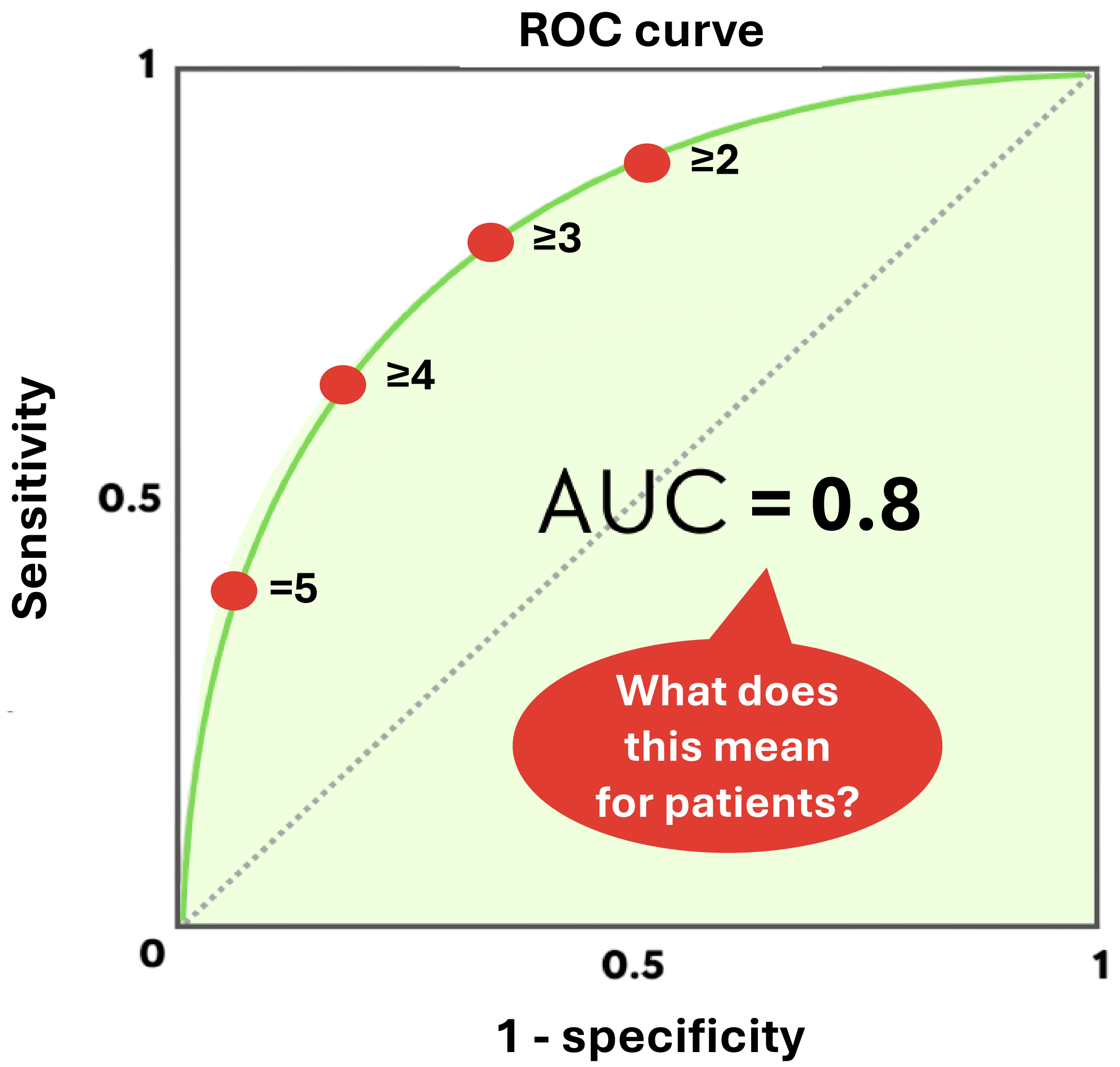
Participating in research should foster your scepticism

Copying from previous studies causes errors to spread like viruses

Scepticism holds us to high standards

BE SCEPTICAL

- A virus infecting diagnostic test accuracy studies
- How accurately can MRI detect brain cancer relative to biopsy using a Likert score?
- AUC averages sensitivity and specificity across thresholds
- Consequences of false positives and false negatives are different



Halligan S, Altman DG, Mallett S: Disadvantages of using the area under the receiver operating characteristic curve to assess imaging tests: A discussion and proposal for an alternative approach. *European Radiology 2015, 25(4):932–939*

PREDEFINE EVERYTHING

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UCL

Why do all this work, even if brief?

- Protocol → CTIMP protocol development tool
- Statistical analysis plan → NIH National Library of Medicine, National Center for Biotechnology Information; ClinicalTrials.gov
- Reporting guidelines → equator network
- Risk of bias tools → LATITUDES network

→ **Focus and predefine** →

- Research question
- Outcomes to answer research question
- Collaborators and their roles

PREDEFINE EVERYTHING

UCL

Studies outcomes can be reshaped to tell a different story that fits the data

1. Not reporting predefined outcomes
2. Changing rank of an outcome
3. Adding new outcomes
4. Changing outcome definitions

- Hiding unsuccessful outcomes
- Highlighting successful outcomes

A 2021 review found 9 out of 67 RCTs reported outcomes exactly as predefined

Changing outcomes is not always bad, it is the covert changing of outcomes that is unacceptable

1. Distorts performance of test or drug
2. Clinical guidelines based on distorted outcomes
3. Patients undergo tests based on distorted evidence

Altman DG, Moher D, Schulz KF: Harms of outcome switching in reports of randomised trials: CONSORT perspective. *BMJ 2017, 356*

Goldacre B, Drysdale H, Dale A, Milosevic I, Slade E, Hartley P, Marston C, Powell-Smith A, Heneghan C, Mahtani KR et al: COMPare: A prospective cohort study correcting and monitoring 58 misreported trials in real time. *Trials 2019, 20*

Kampman JM, Sperna Weiland NH, Hollmann MW, Repping S, Hermanides J: High incidence of outcome switching observed in follow-up publications of randomized controlled trials: Meta-research study. *Journal of Clinical Epidemiology 2021, 137:236–240*

PRACTICE THE ART OF DATA VISUALISATION

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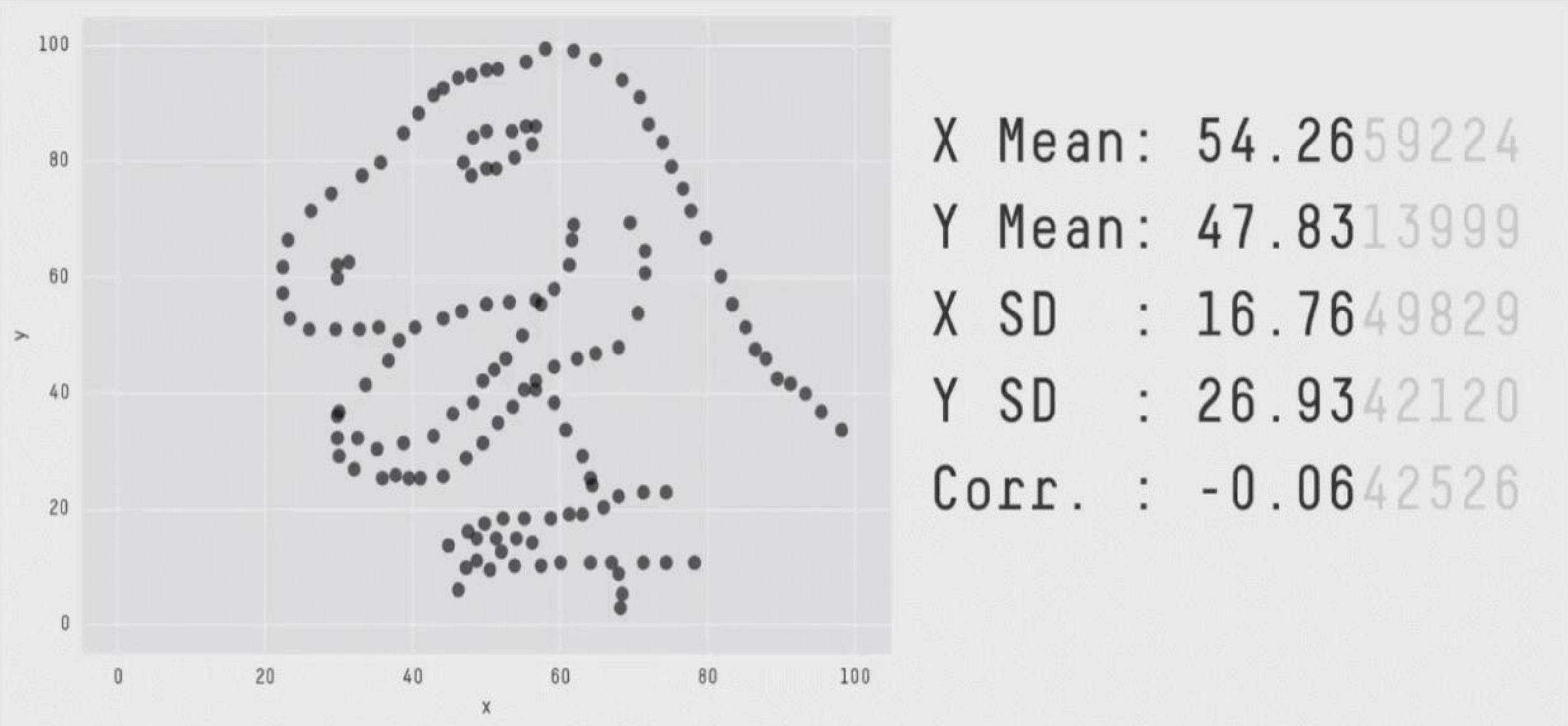
- Allows readers to see beyond summary statistics
- Reveals hidden patterns
- Improves transparency and trust in the data

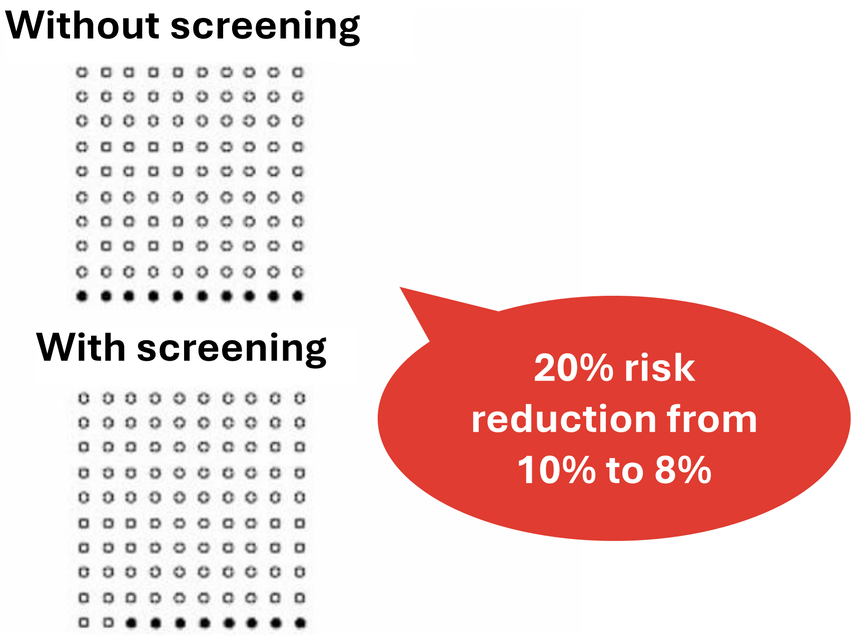


Matejka J, Fitzmaurice G: Same stats, different graphs: Generating datasets with varied appearance and identical statistics through simulated annealing. In.; *2017*

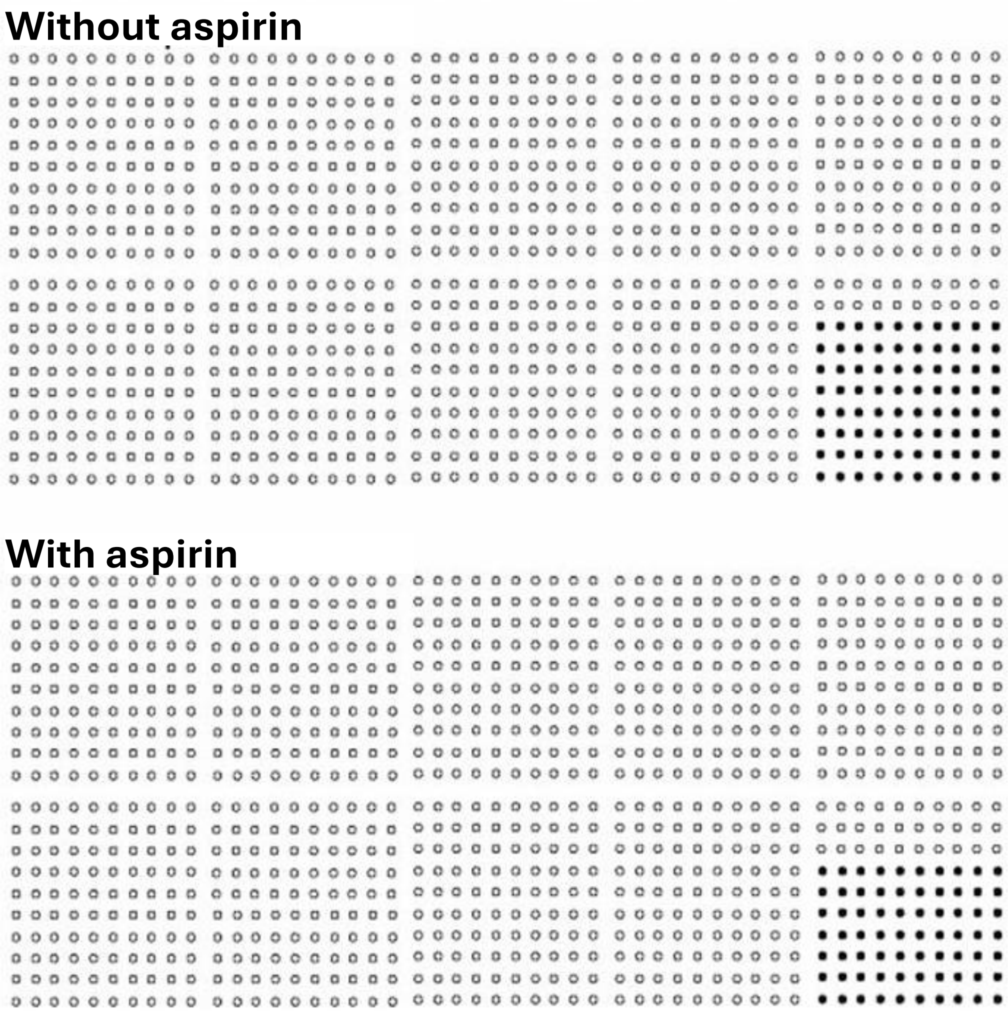
PRACTICE THE ART OF DATA VISUALISATION

- Visualise probability with frequency plots
- Relative risk can be misleading
- Visualise sensitivity and specificity

Black circles show people who will die of cancer with and without screening



For people with arterial disease symptoms, aspirin can reduce the risk of stroke by 13%



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REPORT NUMBERS WITH UNITS

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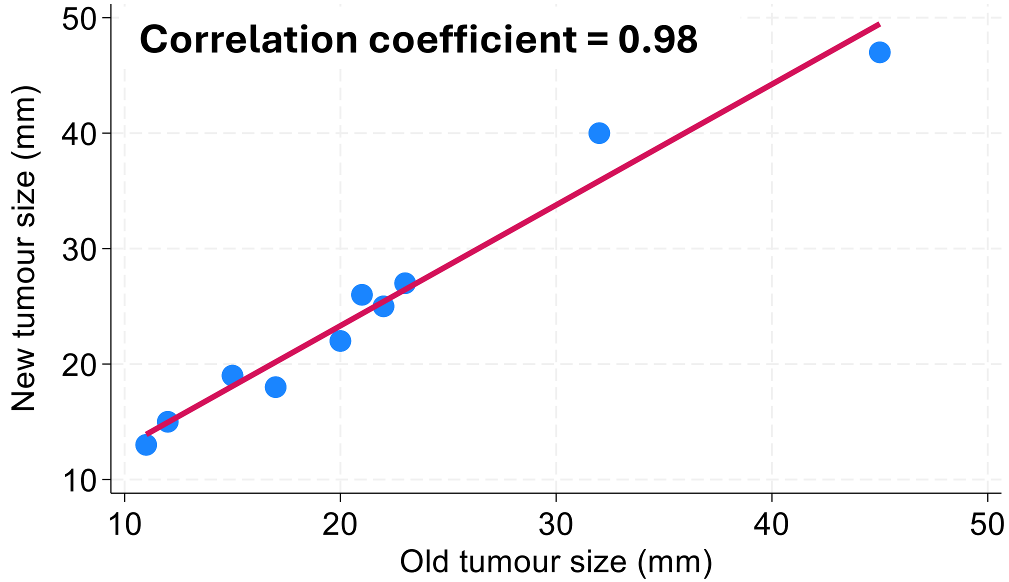
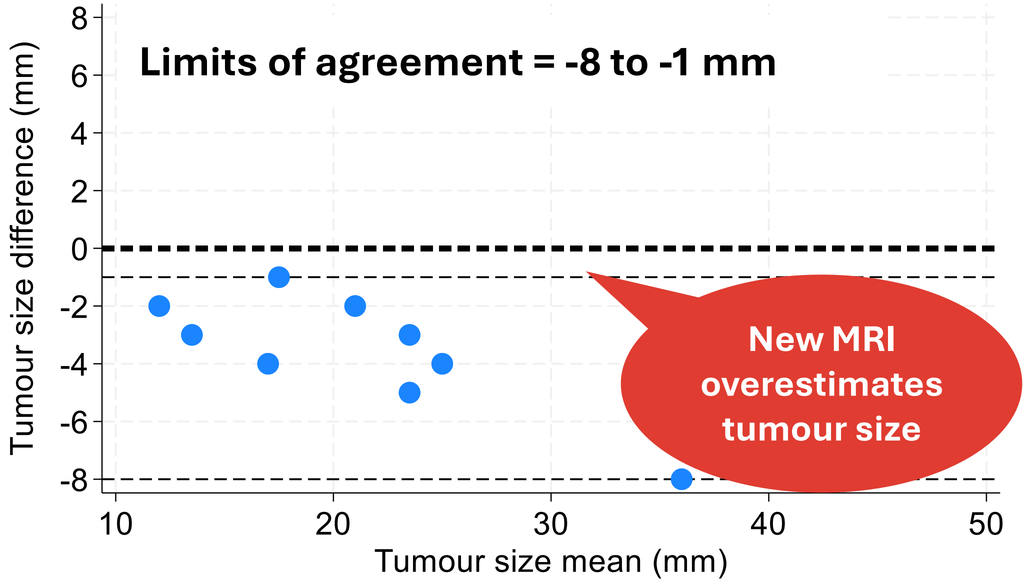
- Correlation coefficients, *p* values etc.
- Does a new fast MRI sequence provide acceptably similar tumour sizes to an older slow sequence
- Is this difference acceptable?
- We need units to understand how numbers relate to the real world
- Units enable decision-making





STATISTICAL SIGNIFICANCE ≠ CLINICAL SIGNIFICANCE

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- Statistical significance defined as $p < 0.05$
- Probability of observing a difference this large, assuming it is not real (mental gymnastics)
- Encourages selective reporting, *p* hacking, and emphasis on “positive” findings
- Leads to adoption of tests and drugs with minimal patient benefit
- Leads to studies not being conducted due to previous “negative” findings
- Is the difference large and precise enough to be clinically significant?

MRI sensitivity = 92%
CT sensitivity = 90%

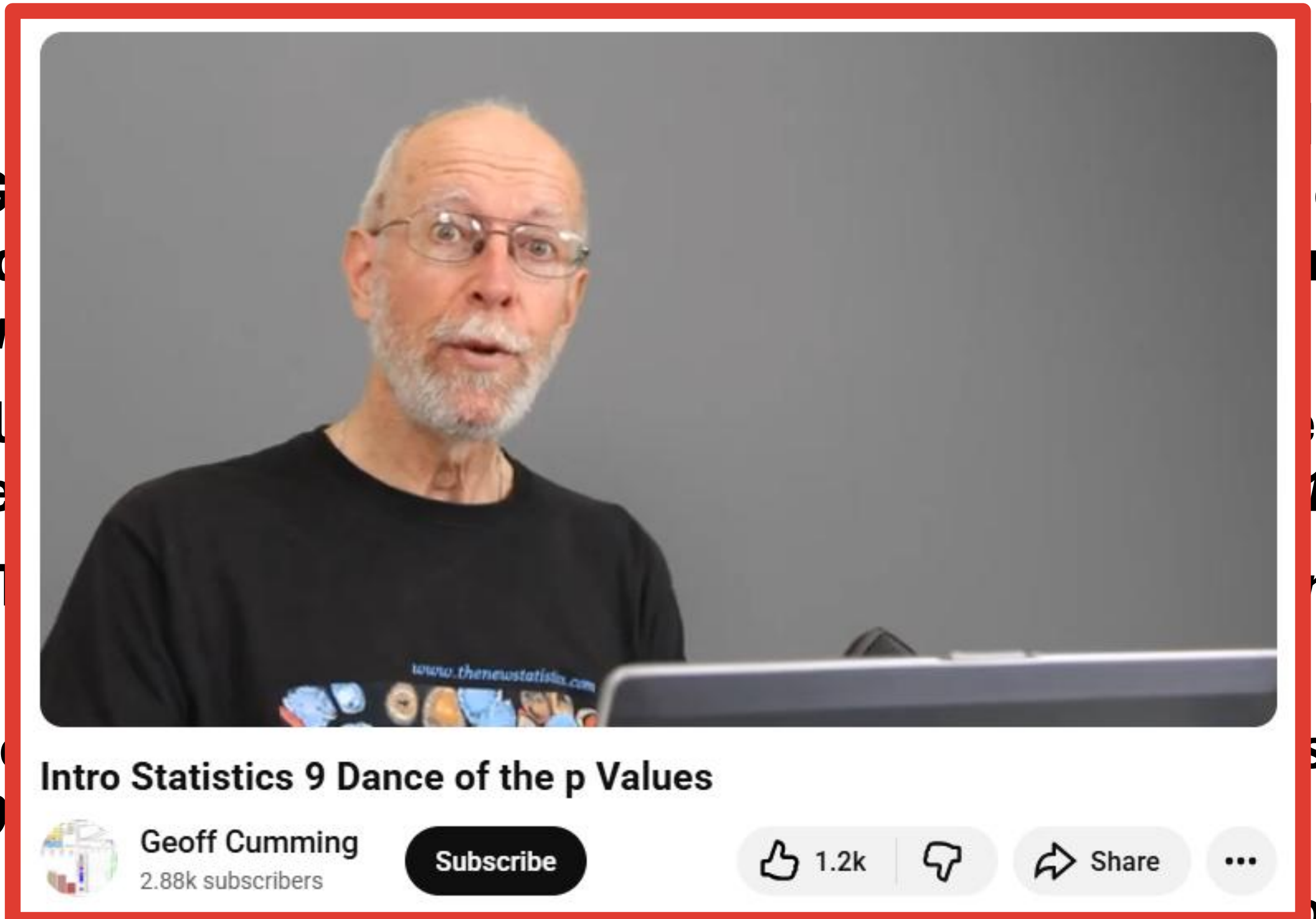
Hypothesis testing

"MRI was significantly more sensitive than CT for detecting brain tumours ($p = 0.02$)"

"MRI sensitivity was 2% (95% CI 0.3% to 3.7%) higher than CT for detecting brain tumours”

“MRI correctly detected 20 more patients with a brain tumour per 1000 tested (95% CI 3 to 37 more patients) than CT”

- Greenland S, … man SN, Altman DG, G… cal tests, P values, confi… rpretations. *European Jou…*
- Head ML, Hol… xtent and consequence… *13(3):e1002106*
- Matthews R: … *nce 2021, 18(2):16–19*
- Cohen J: The … *st 1994, 49(12):997–10…*
- McShane BB, … on statistical significance. *The American Statistician 2019, 73:235–245*



BE HONEST

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1. Be honest during study design
2. Be honest while you conduct your study
3. Be honest about your data
4. Be honest about the clinical significance of your findings

Thinking like a medical statistician

- **Attend medical statistics conferences, meet statisticians, grow your network**
- **Find the leading methodologists/statisticians doing similar studies to yours**
- **Find authors from the risk of bias/reporting guidelines that apply to your study**
- **Find a statistician supervisor for your PhD**
- **Learn to code using statistical software**
- **Use SciHub to access articles**
- **ResearchRabbit for publication linkage**

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